

Three Multiplier Channels and a Lefschetz Identity for the Full Multiplication Lattès Family

Route-A structural certificate HCS-C180

Abstract

Let an integer multiplication map $[m]$ on a complex elliptic curve descend through the quotient by sign to a Lattès map f_m on the sphere. Uniformly over every elliptic modulus, $m \geq 2$, and iterate $n \geq 1$, we resolve the fixed set into regular plus, regular minus, and branch channels. With $a = m^n$ and $h = 1$ for even a , $h = 4$ for odd a , their counts are $((a-1)^2 - h)/2$, $((a+1)^2 - h)/2$, and h , with multipliers $a, -a, a^2$. They give a holomorphic Lefschetz sum equal to one. We derive the exact period ledger, $\zeta_{\text{AM}} = ((1-z)(1-m^2z))^{-1}$, and the Wold model $I_{\mathbb{C}} \oplus S^{(\aleph_0)}$ of the natural Haar Koopman isometry. The theorem is a full Lattès classification but a sharp Route-A stop: its counts are modulus-blind and its canonical operator is noncompact.

Keywords: Lattès map; elliptic torsion; holomorphic Lefschetz formula; Artin–Mazur zeta; Wold decomposition.

中文摘要

对任意复椭圆曲线、整数 $m \geq 2$ 与迭代次数 $n \geq 1$ ，本文完整分类乘法 $[m]$ 经 $\{\pm 1\}$ 商降下的拉泰斯映射不动点。令 $a = m^n$ ；不动类分为正、负与分支三通道，其乘子分别为 $a, -a, a^2$ ，计数中的分支修正由 a 的奇偶性决定。三通道给出恰为一的莱夫谢茨和，并导出精确周期、动力学泽塔与自然哈尔–库普曼等距算子的沃尔德分解。该结果是完整动力学定理，但因计数对模参数失明且算子非紧，它同时构成严格的路线停止定理。

1 Family and fixed-class theorem

Let $E = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ be any complex elliptic curve, $\pi : E \rightarrow E/\{\pm 1\} \simeq \mathbb{P}^1$, and $m \geq 2$. Define f_m by $f_m\pi = \pi[m]$. This classical Lattès construction is prior work; see Milnor [1]. We claim no priority for that construction, torsion periodic points, or the holomorphic Lefschetz formula.

Fix $n \geq 1$, put $a = m^n$, and define $h(a) = 1$ when a is even and $h(a) = 4$ when a is odd.

Theorem 1 (Three-channel census). *For every E, m, n ,*

$$\text{Fix}(f_m^n) = (E[a-1] \cup E[a+1])/\{\pm 1\}.$$

The fixed classes split disjointly as

<i>channel</i>	<i>count</i>	<i>multiplier</i>	<i>origin</i>
<i>regular plus</i>	$N_+ = ((a-1)^2 - h)/2$	$+a$	$E[a-1]$
<i>regular minus</i>	$N_- = ((a+1)^2 - h)/2$	$-a$	$E[a+1]$
<i>branch</i>	$N_{\text{br}} = h$	a^2	<i>common two-torsion</i>

In particular, $\#\text{Fix}(f_m^n) = a^2 + 1$ and

$$\frac{N_+}{1-a} + \frac{N_-}{1+a} + \frac{N_{\text{br}}}{1-a^2} = 1. \quad (1)$$

Proof. The equality $f_m^n\pi(P) = \pi(P)$ is equivalent to $[a]P = P$ or $[a]P = -P$, which gives the torsion union. Its intersection is $E[\gcd(a-1, a+1)]$: it has one element for even a and four for odd a . These are exactly the quotient branch classes. Remove the intersection and pair the remaining points by sign to obtain the counts.

Off the branch locus, π is locally biholomorphic. The chain rule gives $+a$ on the first torsion set. On the second, the identity $D\pi_{-P} = -D\pi_P$ supplies the sign $-a$. At two-torsion a quotient coordinate is the square of a torus coordinate, so the multiplier is a^2 . The total and (1) follow by direct algebra. $\square$

Why the branch channel cannot be merged

The common torsion classes are not double-counted regular points. Their local coordinate is quadratic, so replacing them by either regular channel would assign multiplier $\pm a$ instead of a^2 . The resulting fixed total could still be repaired arithmetically, but the local Lefschetz sum would be wrong. Thus the three-channel statement is strictly stronger than the scalar count $a^2 + 1$ and is the minimal branch-aware invariant of the quotient.

2 Exact periods and zeta

Corollary 2. *The exact-period point count and primitive-cycle count are*

$$P_m(n) = \sum_{d|n} \mu(n/d)(m^{2d} + 1), \quad P_m(n)/n,$$

and

$$\zeta_{\text{AM}}(z) = \exp\left(\sum_{n \geq 1} (m^{2n} + 1) \frac{z^n}{n}\right) = \frac{1}{(1-z)(1-m^2z)}. \quad (2)$$

Proof. Möbius inversion separates exact periods, and exact-period points partition into cycles of size n . Equation (2) is the sum of two geometric logarithmic series. $\square$

3 Natural operator and Wold boundary

Let $\nu = \pi_* \text{Haar}_E$ and let $U_m g = g \circ f_m$ on $L^2(\mathbb{P}^1, \nu)$. Pullback identifies this Hilbert space with the even subspace of $L^2(E)$. For nonzero dual-lattice k , set $c_k = (e_k + e_{-k})/\sqrt{2}$, indexed modulo sign. Then

$$U_m c_k = c_{mk}.$$

Every nonzero $k \in \mathbb{Z}^2$ is uniquely $m^j r$ with $r \notin m\mathbb{Z}^2$, modulo sign. Each primitive r starts a unilateral shift chain, and there are countably many such roots. Constants form the only unitary summand. Therefore

$$U_m \simeq I_{\mathbb{C}} \oplus S^{(\aleph_0)}, \quad U_m^* c_k = \begin{cases} c_{k/m}, & k \in m\mathbb{Z}^2, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

The isometry is proper, noncompact, and outside every finite Schatten class; an ordinary Fredholm determinant $\det(I - zU_m)$ is unavailable. Pullback of an ample line bundle is only a formal alternative: $[m]^* L$ has degree $m^2 \deg L$, so it does not act on one fixed quantization space.

This operator statement uses only pushed-forward Haar measure. It neither chooses a polarization nor identifies Hilbert spaces of different line-bundle degree. Consequently the Wold model is canonical for the frozen measurable system, whereas the line-bundle suggestion is deliberately recorded only as a change-of-space hint.

4 Route-A decision

The exact tuple is

$$(\text{A0_FAIL}, \text{A1_WEAK}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_FORMAL_HINT}).$$

A0 fails because torsion classes have no intrinsic rational-prime labels or prime-power weights. A1 remains weak despite complete orbit counts because no arithmetic labels or target amplitudes occur. The elementary zeta is independent of τ , giving A2 and A3 failure. Equation (3) is canonical but non-Schatten, so A4 is only a formal hint. A0 failure forces `ROUTE_A_REJECTED`.

The scope is `NO_BAD_EULER_OR_ROOT_NUMBER`. No prime table, target divisor, arithmetic local factor, Euler factor, root number, automorphy, Hilbert–Pólya operator, Route-B authorization, external review, or acceptance rate is claimed.

References

- [1] J. Milnor, *On Lattès Maps*, arXiv:math/0402147 (2004), <https://arxiv.org/abs/math/0402147>.